

11.07 Step Functions

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Functions And Models	Read the function form and the context before choosing a graphing or modeling method.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Step Functions

1.

$$f(x) = x^2$$

2.

$$y = 2(x - 3)^2 - 5$$

3.

$$\frac{f(4) - f(1)}{4 - 1}$$

4.

$$f(x) = x^2$$

5.

$$y = 2(x - 3)^2 - 5$$

6.

$$\frac{f(4) - f(1)}{4 - 1}$$

7.

$$f(x) = x^2$$

8.

$$y = 2(x - 3)^2 - 5$$

9.

$$\frac{f(4) - f(1)}{4 - 1}$$

10.

$$f(x) = x^2$$

11.

$$y = 2(x - 3)^2 - 5$$

12.

$$\frac{f(4) - f(1)}{4 - 1}$$

13.

$$f(x) = x^2$$

14.

$$y = 2(x - 3)^2 - 5$$

15.

$$\frac{f(4) - f(1)}{4 - 1}$$

16.

$$f(x) = x^2$$

17.

$$y = 2(x - 3)^2 - 5$$

18.

$$\frac{f(4) - f(1)}{4 - 1}$$

19.

$$f(x) = x^2$$

20.

$$y = 2(x - 3)^2 - 5$$

B. Apply the key formulas

1.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

2.

vertex of $y = a(x - h)^2 + k : (h, k)$

3.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

4.

$$m = \frac{y_2 - y_1}{d_2 - d_1}$$

5.

vertex of $y = a(x - h)^2 + k : (h, k)$

6.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

7.

$$m = \frac{y_2 - y_1}{b_2 - b_1}$$

8.

vertex of $y = a(x - h)^2 + k : (h, k)$

9.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

10.

$$m = \frac{y_2 - y_1}{e_2 - e_1}$$

11.

vertex of $y = a(x - h)^2 + k : (h, k)$

12.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

13.

$$m = \frac{y_2 - y_1}{c_2 - c_1}$$

14.

vertex of $y = a(x - h)^2 + k : (h, k)$

15.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

16.

$$m = \frac{y_2 - y_1}{a_2 - a_1}$$

17.

vertex of $y = a(x - h)^2 + k : (h, k)$

18.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

19.

$$m = \frac{y_2 - y_1}{d_2 - d_1}$$

20.

vertex of $y = a(x - h)^2 + k$: (h, k)